

Ratios, Proportion & Percentages

ANSWER KEY



Simplifying ratios

- 1 Simplify the ratio 18 : 24.
GCD of 18 and 24 is 6: $18 \div 6 : 24 \div 6 = 3 : 4$.
 - 2 Simplify the ratio 45 : 60 : 75.
GCD of 45, 60, 75 is 15: $3 : 4 : 5$.
 - 3 Simplify the ratio 36 : 8.
GCD of 36 and 8 is 4: $9 : 2$.
 - 4 Express "12 boys to 18 girls" as a ratio in simplest form.
 $12 : 18$, and GCD is 6, so $2 : 3$.
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Sharing a quantity in a given ratio

- 5 Share 150 in the ratio 2 : 3.
 $2 + 3 = 5$ parts, each part $= 150 \div 5 = 30$. Shares: 60 and 90.
 - 6 Share 220 among three people in the ratio 3 : 4 : 4.
 $3 + 4 + 4 = 11$ parts, each part $= 220 \div 11 = 20$. Shares: 60, 80, 80.
 - 7 A prize of 900 riyals is shared between Sara and Layla in the ratio 5 : 4. Find each share.
 $5 + 4 = 9$ parts, each part $= 900 \div 9 = 100$. Sara receives 500 riyals; Layla receives 400 riyals.
 - 8 Divide 84 kg of flour in the ratio 1 : 2 : 4.
 $1 + 2 + 4 = 7$ parts, each part $= 84 \div 7 = 12$ kg. Shares: 12 kg, 24 kg, 48 kg.
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Equivalent ratios and proportions

- 9 If $3 : 5 = x : 20$, find x .
 $\frac{3}{5} = \frac{x}{20}$, so $x = \frac{3 \times 20}{5} = 12$.
- 10 If $a : 7 = 12 : 21$, find a .
 $\frac{12}{21} = \frac{a}{7}$, so $a = 4$.
- 11 A map has scale 1 : 50,000. A distance of 3 cm on the map represents how many meters in reality?
 $3 \times 50,000 = 150,000$ cm $= 1,500$ m.

12 Solve for y : $\frac{4}{4 \times 63} = \frac{y}{63}$.
 $y = \frac{4}{9} = 28$.

Percent of a number

13 Find 35% of 240.
 $\frac{35}{100} \times 240 = 84$.

14 Find 12% of 150.
 $\frac{12}{100} \times 150 = 18$.

15 What is 8% of 650?
 $\frac{8}{100} \times 650 = 52$.

16 Find 150% of 40.
 $\frac{150}{100} \times 40 = 60$.

Converting between fractions, decimals and percentages

17 Write $\frac{3}{8}$ as a percent.
 $\frac{3}{8} = 0.375 = 37.5\%$.

18 Write 0.45 as a fraction in lowest terms.
 $0.45 = \frac{45}{100} = \frac{9}{20}$.

19 Write 62.5% as a fraction in lowest terms.
 $62.5\% = \frac{625}{1000} = \frac{5}{8}$.

20 Write $\frac{7}{40}$ as a decimal.
 $\frac{7}{40} = 0.175$.

Percentage increase and decrease

21 Increase 80 by 25%.
 $80 \times 1.25 = 100$.

22 Decrease 150 by 20%.
 $150 \times 0.8 = 120$.

23 A salary of 4,000 riyals increases by 6%. Find the new salary.
 $4,000 \times 1.06 = 4,240$ riyals.

- 24 A population of 2,500 decreases by 12%. Find the new population.
 $2,500 \times 0.88 = 2,200$.
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Finding percentage change and original values

- 25 A price rises from 80 to 92. Find the percentage increase.
Change = $92 - 80 = 12$. Percent increase = $\frac{12}{80} \times 100\% = 15\%$.
- 26 A quantity falls from 250 to 200. Find the percentage decrease.
Change = $250 - 200 = 50$. Percent decrease = $\frac{50}{250} \times 100\% = 20\%$.
- 27 After a 20% discount, a jacket costs 96 riyals. Find the original price.
Original $\times 0.8 = 96$, so original = $96 \div 0.8 = 120$ riyals.
- 28 After a 15% increase, a value becomes 138. Find the original value.
Original $\times 1.15 = 138$, so original = $138 \div 1.15 = 120$.
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Successive percentage changes

- 29 A price of 100 rises 10%, then falls 10%. Find the final price.
 $100 \times 1.1 \times 0.9 = 99$ — one riyal less than the start, since the second change applies to a larger base.
- 30 A stock worth 200 riyals rises 20% one year, then rises a further 25% the next year. Find the final value.
 $200 \times 1.2 \times 1.25 = 300$ riyals.
- 31 A quantity decreases by 10%, then decreases by a further 10%. Find the overall percentage decrease from the original amount.
 $100 \times 0.9 \times 0.9 = 81$, a fall of 19 from 100 — an overall decrease of 19% (not 20%).
- 32 A price of 400 riyals increases by 5%, then decreases by 5%. Find the final price, and state whether it equals the starting price.
 $400 \times 1.05 \times 0.95 = 399$ riyals — one riyal less than the original 400, so it is not equal.
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Direct proportion

- 33 y is directly proportional to x , and $y = 15$ when $x = 3$. Find y when $x = 8$.
 $y = kx$ with $k = 15 \div 3 = 5$. When $x = 8$: $y = 5 \times 8 = 40$.
- 34 4 kg of apples cost 28 riyals. Find the cost of 7 kg at the same rate.
Cost per kg = $28 \div 4 = 7$ riyals. Cost of 7 kg = $7 \times 7 = 49$ riyals.

- 35** A car uses 8 liters of fuel to travel 96 km. How far can it travel on 20 liters at the same rate?
Rate = $96 \div 8 = 12$ km per liter. Distance = $20 \times 12 = 240$ km.
- 36** 6 identical pens cost 27 riyals. Find the cost of 10 pens.
Cost per pen = $27 \div 6 = 4.5$ riyals. Cost of 10 pens = $10 \times 4.5 = 45$ riyals.
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Inverse proportion

- 37** 4 workers can complete a job in 15 days. How many days would 6 workers take, working at the same rate?
Total work = $4 \times 15 = 60$ worker-days. With 6 workers: $60 \div 6 = 10$ days.
- 38** A car covers a fixed distance in 5 hours at 60 km/h. How long would the same trip take at 75 km/h?
Distance = $60 \times 5 = 300$ km. Time at 75 km/h: $300 \div 75 = 4$ hours.
- 39** 8 taps fill a pool in 12 hours. How long would 3 taps take, running at the same rate?
Total work = $8 \times 12 = 96$ tap-hours. With 3 taps: $96 \div 3 = 32$ hours.
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Best value and unit price comparisons

- 40** A 3 kg bag of rice costs 27 riyals, and a 5 kg bag costs 42.50 riyals. Which is the better value per kilogram?
3 kg bag: $27 \div 3 = 9$ riyals/kg. 5 kg bag: $42.5 \div 5 = 8.5$ riyals/kg. The 5 kg bag is better value.
- 41** Store A sells 6 pens for 15 riyals; Store B sells 8 pens for 19.20 riyals. Which store offers the lower price per pen?
Store A: $15 \div 6 = 2.5$ riyals/pen. Store B: $19.2 \div 8 = 2.4$ riyals/pen. Store B has the lower price per pen.
- 42** A 2-liter bottle of juice costs 14 riyals. Find the price per 100 mL.
2 liters = 20 portions of 100 mL, so price per 100 mL = $14 \div 20 = 0.7$ riyals.
- 43** Which is the better deal: 12 eggs for 9 riyals, or 18 eggs for 12.60 riyals?
12 eggs: $9 \div 12 = 0.75$ riyals/egg. 18 eggs: $12.6 \div 18 = 0.70$ riyals/egg. The 18-egg pack is the better deal.
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Word problems

- 44** A recipe requires flour and sugar in the ratio 5 : 2. If 350 g of flour is used, how much sugar is needed?
Scale factor = $350 \div 5 = 70$. Sugar = $2 \times 70 = 140$ g.

- 45 A shop raises the price of a shirt from 60 to 75 riyals, then during a sale reduces the new price by 20%. Find the sale price.
New price after the sale: $75 \times 0.8 = 60$ riyals — back to the original price.
- 46 In a class of 40 students, 60% are girls. Then 4 more girls join the class. What percent of the class is now girls, to the nearest percent?
Girls initially $= 0.6 \times 40 = 24$; boys $= 16$. After 4 more girls: 28 girls out of 44 students.
 $\frac{28}{44} \times 100\% \approx 64\%$.
- 47 A dealership reduces the price of a car by 15% to 25,500 riyals. Find the original price.
Original $\times 0.85 = 25,500$, so original $= 25,500 \div 0.85 = 30,000$ riyals.
- 48 Ahmed invests 8,000 riyals in an account that grows by 5% each year. Find its value after 2 years.
 $8,000 \times 1.05^2 = 8,000 \times 1.1025 = 8,820$ riyals.
- 49 A map has scale 1 : 25,000. Two towns are 8.4 cm apart on the map. Find the actual distance between them in kilometers.
 $8.4 \times 25,000 = 210,000$ cm $= 2,100$ m $= 2.1$ km.
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MCQ — Simplifying & writing ratios

- 50 Simplify the ratio 42 : 56.

a. 3 : 4 ✓

b. 6 : 8

c. 7 : 8

d. 3 : 8

Divide both terms by their GCD, 14: $42 \div 14 : 56 \div 14 = 3 : 4$.

- 51 Simplify 18 : 45.

a. 3 : 5

b. 2 : 9

c. 2 : 5 ✓

d. 6 : 15

GCD of 18 and 45 is 9: $18 \div 9 : 45 \div 9 = 2 : 5$.

52 A recipe uses 250 g flour and 2 eggs. Write the ratio of flour (in grams) to eggs in simplest form.

a. 1 : 125

b. 250 : 2

c. 125 : 1 ✓

d. 2 : 250

250 : 2 simplifies by dividing both by 2: 125 : 1.

53 Express 0.75 : 2 as a ratio of whole numbers in simplest form.

a. 3 : 4

b. 3 : 8 ✓

c. 75 : 2

d. 75 : 200

Multiply both terms by 4 to clear the decimal: 3 : 8.

MCQ — Sharing in a given ratio

54 The bar chart shows the ratio in which a total of 84 is to be shared. Find the larger share.

a. 36

b. 21

c. 48 ✓

d. 63

Total parts = $3 + 4 = 7$; each part = $84 \div 7 = 12$. Larger share = $4 \times 12 = 48$.

55 Share 260 SAR in the ratio 2 : 3 : 5. Find the smallest share.

- a. 78 SAR
- b. 52 SAR ✓
- c. 130 SAR
- d. 65 SAR

Total parts = $2 + 3 + 5 = 10$; each part = $260 \div 10 = 26$. Smallest share = $2 \times 26 = 52$.

56 Two business partners share a profit of 18,000 SAR in the ratio 5 : 7. Find the larger share.

- a. 10,500 SAR ✓
- b. 9,000 SAR
- c. 7,500 SAR
- d. 12,857 SAR

Each part = $18,000 \div 12 = 1,500$. Larger share = $7 \times 1,500 = 10,500$ SAR.

57 A line of length 36 cm is divided in the ratio 4 : 5. Find the length of the shorter piece.

- a. 16 ✓
- b. 4
- c. 9
- d. 20

Total parts = $4 + 5 = 9$; each part = $36 \div 9 = 4$ cm. Shorter piece = $4 \times 4 = 16$ cm.

58 A class of 40 students has boys and girls in the ratio 3 : 5. How many boys are there?

- a. 25
- b. 8
- c. 24
- d. 15 ✓

Total parts = $3 + 5 = 8$; each part = $40 \div 8 = 5$. Boys = $3 \times 5 = 15$.

MCQ — Equivalent ratios & scale

59 If $a : b = 3 : 5$ and $a = 18$, find b .

a. 30 ✓

b. 15

c. 10.8

d. 24

Scale factor $= 18 \div 3 = 6$. $b = 5 \times 6 = 30$.

60 A map has scale $1 : 50,000$. A distance of 3 cm on the map represents how many km in reality?

a. 15 km

b. 1.5 km ✓

c. 150 km

d. 0.15 km

$3 \times 50,000 = 150,000 \text{ cm} = 1,500 \text{ m} = 1.5 \text{ km}$.

61 Which ratio is equivalent to $4 : 9$?

a. $9 : 4$

b. $20 : 45$ ✓

c. $8 : 16$

d. $16 : 37$

$4 : 9$ scaled by 5 gives $20 : 45$.

- 62 In a photo, a model that is 1.8 m tall appears 6 cm high. Using the same scale, how tall (in m) is a building that appears 40 cm high in the photo?
- a. 4.8
 - b. 24
 - c. 7.2
 - d. 12 ✓

Scale: 6 cm represents 1.8 m, so 1 cm represents 0.3 m. $40 \times 0.3 = 12$ m.

MCQ — Percent of a number

- 63 Find 35% of 260.
- a. 86
 - b. 95
 - c. 78
 - d. 91 ✓

$$0.35 \times 260 = 91.$$

- 64 Find 12% of 450.
- a. 62
 - b. 48
 - c. 54 ✓
 - d. 45

$$0.12 \times 450 = 54.$$

65 The bar chart shows four students' quiz scores out of 80. Which score is exactly 75%?

- a. 68
- b. 64
- c. 60 ✓
- d. 56

$$75\% \text{ of } 80 = 0.75 \times 80 = 60.$$

66 The bar chart shows monthly rent for four apartments. 8% of which apartment's rent equals 200 SAR?

- a. Apt. 2
- b. Apt. 1
- c. Apt. 3 ✓
- d. Apt. 4

$$0.08 \times 2,500 = 200 \text{ SAR, matching Apartment 3's rent of 2,500 SAR.}$$

67 150% of a number is 300. Find the number.

- a. 250
- b. 450
- c. 200 ✓
- d. 150

$$1.5 \times x = 300 \Rightarrow x = 300 \div 1.5 = 200.$$

MCQ — Fraction, decimal & percent conversion

68 Write $\frac{7}{20}$ as a percent.

a. 35% ✓

b. 3.5%

c. 7%

d. 70%

$$\frac{7}{20} = \frac{35}{100} = 35\%.$$

69 Write 0.06 as a percent.

a. 60%

b. 6% ✓

c. 0.06%

d. 0.6%

Move the decimal point two places right: $0.06 = 6\%$.

70 Write 85% as a fraction in simplest form.

a. $\frac{17}{20}$ ✓

b. $\frac{8}{5}$

c. $\frac{17}{25}$

d. $\frac{85}{100}$

$$85\% = \frac{85}{100}, \text{ which simplifies (divide by 5) to } \frac{17}{20}.$$

71 Which of these is largest: $\frac{3}{8}$, 0.4, or 37%?

a. 0.4 ✓

b. They are all equal

c. $\frac{3}{8}$

d. 37%

$$\frac{3}{8} = 0.375 = 37.5\%, \text{ and } 37\% = 0.37. \text{ So } 0.4 \text{ is the largest.}$$

MCQ — Percentage increase & decrease

72 Increase 80 by 25%.

- a. 105
- b. 20
- c. 100 ✓
- d. 95

$$80 \times 1.25 = 100.$$

73 Decrease 150 by 12%.

- a. 138
- b. 128
- c. 132 ✓
- d. 18

$$150 \times 0.88 = 132.$$

74 The bar chart shows a salary before and after a raise. What percentage increase does it represent?

- a. 15%
- b. 500%
- c. 20% ✓
- d. 25%

$$\text{Increase} = 6,000 - 5,000 = 1,000. \text{ Percentage increase} = \frac{1,000}{5,000} \times 100\% = 20\%.$$

75 A shirt costs 120 SAR after a 20% discount. Find the original price.

a. 150 SAR ✓

b. 144 SAR

c. 100 SAR

d. 140 SAR

$$120 = 0.8 \times \text{original} \Rightarrow \text{original} = 120 \div 0.8 = 150 \text{ SAR.}$$

76 Find the percentage change from 40 to 34.

a. 17.6% decrease

b. 15% increase

c. 6% decrease

d. 15% decrease ✓

$$\text{Change} = 34 - 40 = -6. \text{ Percentage change} = \frac{-6}{40} \times 100\% = -15\%, \text{ a 15\% decrease.}$$

MCQ — Finding the original value

77 After a 15% increase, a price is 92 SAR. Find the original price.

a. 80 SAR ✓

b. 78.20 SAR

c. 88 SAR

d. 107 SAR

$$92 = 1.15 \times \text{original} \Rightarrow \text{original} = 92 \div 1.15 = 80 \text{ SAR.}$$

78 After a 30% discount, an item costs 56 SAR. Find the original price.

a. 72.80 SAR

b. 80 SAR ✓

c. 86 SAR

d. 74.67 SAR

$$56 = 0.7 \times \text{original} \Rightarrow \text{original} = 56 \div 0.7 = 80 \text{ SAR.}$$

79 A population grew by 8% to reach 27,000. Find the population before the growth.

a. 25,000 ✓

b. 26,000

c. 29,160

d. 24,840

$$27,000 = 1.08 \times \text{original} \Rightarrow \text{original} = 27,000 \div 1.08 = 25,000.$$

80 After including a 15% service charge, a restaurant bill is 138 SAR. Find the bill before the charge.

a. 158.70 SAR

b. 120 SAR ✓

c. 123 SAR

d. 117.30 SAR

$$138 = 1.15 \times \text{original} \Rightarrow \text{original} = 138 \div 1.15 = 120 \text{ SAR.}$$

MCQ — Successive percentage changes

81 A price of 200 rises 10%, then rises another 10%. Find the final price.

- a. 220
- b. 240
- c. 244
- d. 242 ✓

$$200 \times 1.1 \times 1.1 = 242.$$

82 A price of 500 falls 20%, then rises 20%. Find the final price.

- a. 460
- b. 480 ✓
- c. 500
- d. 520

$$500 \times 0.8 \times 1.2 = 480 \text{ — a rise then fall of the same percentage never returns to the start.}$$

83 The bar chart shows a stock's value at the start and after rising 50% in year one. It then falls 50% in year two. Find its final value.

- a. 500
- b. 900
- c. 1,000
- d. 750 ✓

$$1,000 \times 1.5 \times 0.5 = 750.$$

MCQ — Direct proportion

- 84 If 5 workers can build a wall in a task requiring 200 worker-hours total, and pay is directly proportional to hours worked, how much does one worker earn (in SAR) for 8 hours at a rate of 25 SAR/hour?
- a. 40
 - b. 200 ✓
 - c. 225
 - d. 175

$8 \times 25 = 200$ SAR — direct proportion between hours worked and pay.

- 85 12 pens cost 30 SAR. At the same rate, find the cost of 20 pens.
- a. 45 SAR
 - b. 60 SAR
 - c. 40 SAR
 - d. 50 SAR ✓

Cost per pen = $30 \div 12 = 2.5$ SAR. 20 pens cost $20 \times 2.5 = 50$ SAR.

- 86 The graph shows the direct proportion between the number of hours x worked and pay y in SAR, $y = 15x$. How much is earned for 6 hours?
- a. 105
 - b. 90 ✓
 - c. 75
 - d. 21

$y = 15 \times 6 = 90$ SAR.

- 87 A car uses 8 liters of fuel to travel 96 km. At the same rate, how far can it travel on 20 liters?
- a. 200
 - b. 260
 - c. 240 ✓
 - d. 180

Rate = $96 \div 8 = 12$ km per liter. $20 \times 12 = 240$ km.

MCQ — Inverse proportion

88 6 workers can finish a job in 10 days. How many days would 4 workers take (same rate each)?

a. 15 ✓

b. 6.67

c. 24

d. 12

Total worker-days = $6 \times 10 = 60$. With 4 workers: $60 \div 4 = 15$ days.

89 If y is inversely proportional to x and $y = 8$ when $x = 3$, find y when $x = 12$.

a. 0.5

b. 32

c. 4

d. 2 ✓

$xy = 3 \times 8 = 24$ (constant). When $x = 12$: $y = 24 \div 12 = 2$.

90 A tank is filled by 5 identical pumps in 12 hours. How long would 3 of the same pumps take?

a. 15

b. 20 ✓

c. 7.2

d. 4

Total pump-hours = $5 \times 12 = 60$. With 3 pumps: $60 \div 3 = 20$ hours.

MCQ — Best value & unit price

91 The bar chart shows the price per 100 g for three brands of coffee. Which brand offers the best value?

a. Brand B ✓

b. All are equal

c. Brand C

d. Brand A

The lowest price per 100 g is the best value — Brand B at 9 SAR/100g beats Brand A (12) and Brand C (11).

92 The bar chart shows the total price of two cereal boxes of different sizes: 400 g and 600 g. Which has the lower price per 100 g?

a. Cannot be determined

b. They are equal

c. The 600 g box ✓

d. The 400 g box

400 g: $12 \div 4 = 3$ SAR/100g. 600 g: $16.50 \div 6 = 2.75$ SAR/100g. The 600 g box is cheaper per 100 g.

93 Find the unit price (per kg) if 3.5 kg of rice costs 28 SAR.

a. 7 SAR/kg

b. 8 SAR/kg ✓

c. 9.80 SAR/kg

d. 3.5 SAR/kg

$28 \div 3.5 = 8$ SAR per kg.

94 A 2 L bottle of juice costs 18 SAR. Find the price per 100 mL.

a. 9 SAR

b. 0.18 SAR

c. 1.80 SAR

d. 0.90 SAR ✓

2 L = 2,000 mL. $18 \div 2,000 \times 100 = 0.90$ SAR per 100 mL.

MCQ — Word problems

95 A car travels 240 km using 20 L of fuel. Find its fuel economy in km per liter.

- a. 14
- b. 10
- c. 20
- d. 12 ✓

$$240 \div 20 = 12 \text{ km/L.}$$

96 The bar chart shows subject preference percentages from a survey of 250 students. How many students prefer math?

- a. 60
- b. 90
- c. 110
- d. 100 ✓

$$0.40 \times 250 = 100 \text{ students.}$$

97 A recipe for 4 people needs 2.5 cups of rice. How many cups are needed for 10 people?

- a. 6.25 cups ✓
- b. 6 cups
- c. 5 cups
- d. 25 cups

$$\text{Rice per person} = 2.5 \div 4 = 0.625. \text{ For 10 people: } 10 \times 0.625 = 6.25 \text{ cups.}$$

98 A company's revenue rose from 2.4 million to 3.0 million SAR. Find the percentage increase.

a. 30%

b. 25% ✓

c. 20%

d. 60%

Increase = $3.0 - 2.4 = 0.6$ million. Percentage = $\frac{0.6}{2.4} \times 100\% = 25\%$.

99 A map scale is 1 : 200,000. Two cities are 8.5 cm apart on the map. Find the real distance in km.

a. 1.7

b. 34

c. 17 ✓

d. 8.5

$8.5 \times 200,000 = 1,700,000$ cm = 17,000 m = 17 km.